Qihang Jin USTC

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RESEARCH INTERESTS

NeuroAI, computational neuroscience, multimodal neural–behavior modeling, representation geometry, and mechanistic interpretability.

EDUCATION

University of Science and Technology of China, Hefei, China — Aug. 2024 – Jun. 2027
Master of Engineering in Electronic and Information Engineering

- GPA: 3.66/4.3; First-Class Graduate Scholarship, 2025
- **Relevant Courses:** Neurobiology for Brain-Inspired AI, Computer Vision, and Advanced Artificial Intelligence

Chang'an University, Xi'an, China — Aug. 2018 – Jul. 2022
Bachelor of Engineering in Vehicle Engineering

SELECTED PUBLICATIONS

Qihang Jin, Hongbing Luo, Wenjun Lv, Kun Li, Jian Di, Pengfei Li. Calibrated Task-Routed LoRA Banks: Disentangling Routing Mode and Adapter-Birth Initialization for Continual Image Restoration. *Under review at ACCV 2026*.

Xinming Dai*, **Qihang Jin***, Tianshu Tan, Baiyuan Chen, Hanrui Lyu, Lenny Aharon, Matthew R Whiteway, Liam Paninski, Yizi Zhang. SABLE: Sparse-View Interpretable 3D Animal Behavior Representations for Neural Encoding and Decoding. *Under review at NeurIPS 2026*.

Qinan Zhang*, Xinyu Wang*, **Qihang Jin***, Tianyuan Wang, Jian Di, Hongbing Luo, Pengfei Li, Wenjun Lv, Yu Kang. Towards Reliable VLM Judges: State-Conditional Invariance and Presentation-Aware Diagnostics. *Under review at NeurIPS 2026*.

Qinan Zhang*, **Qihang Jin***. GridWM-Judge: Evaluating Vision-Language Model Judges in Grid Worlds via World Model Deficits. *ICLR 2026 Workshop on World Models*. [OpenReview].

* Equal contribution.

ADDITIONAL PUBLICATIONS

Qihang Jin*, Enze Ge*, Yuhang Xie*, Hongying Luo, Junhao Song, Ziqian Bi, Chia Xin Liang, Jibin Guan, Joe Yeong, Junfeng Hao. Multimodal Representation Learning and Fusion. *arXiv*, Jun. 2025. [arXiv].

Enze Ge, **Qihang Jin**, Yuhang Xie, Hongying Luo, Tianyang Wang, Ziqian Bi, Benji Peng, Junfeng Hao, Junhao Song. From Memory to Alignment: A Comprehensive Review of Large Language Model Optimization. *TechRxiv*, Oct. 2025. [TechRxiv].

Peili Cao, Hong Ling, Hao Guo, JunHao Wang, **Qihang Jin**, Shijie Guo, *et al.* Effect of Brain-Computer Interface on Limb Motor Function after Intracerebral Hemorrhage in Basal Ganglia and Its Rehabilitation Mechanism. *Research Square*, Jul. 2025. [Link].

RESEARCH EXPERIENCE

Research Assistant, Columbia University — *Liam Paninski Lab* — Dec. 2025 – present

SABLE: Sparse-View 3D Animal Behavior Representation

- Co-developed SABLE's cross-view geometry-initialization and camera-calibration pipeline, converting monocular depth estimates into pseudo-point clouds and aligning sparse views through keypoint-assisted rigid registration; developed point-cloud visualizations for reconstruction diagnosis.
- Designed controlled ablations over full 28-keypoint versus paws-and-nose supervision and fixed versus ground-truth camera parameters to diagnose cross-view fragmentation on Cheese3D.

Longitudinal IBL Neural–Behavior Analysis

- Performed initial quality control and exploratory analysis across 60 longitudinal IBL sessions, integrating behavioral trials, multi-probe neural recordings, camera motion energy, and movement features.
- Reproduced the POSSM baseline on the chronic dataset, providing a reference point for subsequent cross-session analyses of neural and behavioral variability.

Same Estimate, Different Update? | *First-Author Research Project* Aug. 2026 – present

- Investigating whether task-trained recurrent networks maintaining the same decoded content can nonetheless implement different observation-response laws for future sensory evidence.
- Designed a causal analysis framework combining mechanism-known calibration systems, history-matched counterfactual observation probes, and content-preserving hidden-state interventions to identify and validate low-rank sensory-revision control subspaces in task-trained recurrent networks.
- Developed a planned reliability-OOD evaluation framework testing whether operator mismatch from a circular Bayesian reference predicts signed over- or under-correction — with matched-model controls, cross-architecture replication (RNN/GRU/LSTM), and a two-stage impair+rescue falsification protocol.

Calibrated Task-Routed LoRA Banks | *First-Author Research Project, USTC* Mar. 2026 – Jul. 2026

- Conceived and led the project end-to-end, including problem formulation, LoRA-bank implementation, controlled experiments, visualization, and manuscript writing, separating routing mode from adapter-birth initialization in continual image restoration.
- Designed a 2×2 routing–initialization study with multi-seed, reverse-chain, cross-backbone, and seven-task controls; established zero old-task PSNR drop and a $+0.54$–$+0.88$ dB plasticity gain from cumulative routing.
- Proposed row-normalized adapter birth and a scale-leakage diagnostic that reduced ρ from $+0.179$ to $+0.019$; showed that learned route prediction, rather than initialization, dominates blind-deployment error.

Reliable Evaluation of Vision–Language Model Judges | *Self-Initiated Research Project, USTC* Oct. 2025 – Jul. 2026

- Originated a research agenda on VLM-judge reliability, formulating state-conditional invariance, presentation-aware diagnostics, and world-model deficits as complementary failure modes.
- Designed the evaluation metrics, benchmarks, and diagnostic protocols and built the initial experimental codebase, which collaborators subsequently expanded across broader models and evaluation settings.
- The resulting research agenda produced two papers: a co-first-author NeurIPS 2026 submission and an ICLR 2026 World Models Workshop paper.

EARLIER WORK

Integrated UAV Inspection Training Platform | *AirSim+UE4.27* Sept. 2024 – Dec. 2025

Automotive Performance Data Processing | *MATLAB/GUI* Sept. 2021 – Jun. 2022

License Plate Recognition (CNN) | *Project Leader* Mar. 2020 – Aug. 2020

Adjustable Limiting and Fixing Device for Automated Machining Aug. 2019 – Jul. 2021

- *CN 110480545 B*. Filed Aug. 23, 2019. Issued Jul. 9, 2021.

SKILLS

ML / DL: Python, PyTorch, Hugging Face, scikit-learn, NumPy, Pandas

Tools / Systems: Git, LaTeX, Linux

Languages: Chinese (Native), English